

# JULFIKER IBN HAIDER

Mechatronics Engineer | Embedded, IoT & Mechanical Design Engineer

Portfolio

## PROFILE

A Mechatronics & Industrial Engineering graduate from CUET with hands-on experience in embedded systems, IIoT product development, and industrial automation. Currently working at **Southern IoT** as an Implementation Lead & Hardware Designer, spearheading end-to-end IIoT deployments for industrial and healthcare sectors. Skilled in firmware development, LPWAN communication, RS485/Modbus protocols, CNC machine fabrication, and IoT cloud infrastructure. Strong leadership background through CUET's core technical robotics club.

## EDUCATION

### B.Sc. Mechatronics & Industrial Eng.

CUET

2020 – 2025 | CGPA: 3.10

### HSC – Science

Tamirul Millat Kamil Madrasah

2017 – 2019 | GPA: 5.00

### SSC – Science

Chaprashirhat Fazil Madrasah

2012 – 2017 | GPA: 4.78

## WORK EXPERIENCE

**EMBEDDED SYSTEMS & IOT ENGINEER**
**Southern IoT**
2025 – Present

*Implementation Lead · Hardware Designer · Firmware Developer*

- Leading hardware design and firmware development for IIoT products deployed across industrial sites.
- Engineered data acquisition modules using RAK3173 (LoRaWAN) and ESP32/ESP8266 with RS485/RS232 (Modbus RTU/ASCII) communication.
- Designed and fabricated a CNC Router for precision workpiece cutting and a Laser CNC Machine for engraving operations.
- Developed an Automatic Wire Bending Machine, automating wire-form production and improving repeatability.
- Established a "Standard Installation Method" toward a One-Tap Installation philosophy, cutting deployment time significantly.
- Led industrial enclosure design balancing ruggedness, cost-efficiency, and brand aesthetics.

**GENERAL SECRETARY**
Robo Mechatronics Association (RMA), CUET
2022 – 2024

- Served as Executive Member and team lead for Expeditious-2023.
- Organized workshops on IoT, Robotics, and Microcontrollers for junior batches at CUET and partner institutions.

## CONTACT


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## CORE SKILLS

- Embedded Systems
- IoT / IIoT Development
- Firmware (C/C++/Python)
- RS485 / Modbus Protocols
- PCB & Enclosure Design
- CNC Machine Building
- ROS 2 / Robotics
- PLC / Industrial Automation

## TECHNOLOGIES

ESP32/ESP8266 · RAK3112/RAK3172 (LORAWAN) · Raspberry Pi 4/5 · MQTT · Node-RED · InfluxDB · Grafana · Modbus RTU/ASCII · ROS 2 · SolidWorks · TIA Portal · MATLAB · Proteus · Linux · FreeRtos · GRBL / G-code

## LANGUEAGE

Bangla ★★★★★

English ★★★★

Arabic ★★

## RELEVANT COURSEWORK

Microcontroller & Interfacing · Control Systems · Power Electronics · Machine Learning · Sensor & Instrumentation · Digital Logic Design · Industrial Automation · FreeRtos · GRBL / G-code · CIM · CAD/CAM · Robotics

## PROJECTS

**★ CORE PRODUCT · SOUTHERN IOT**

### CAREFLOW V1 – INDUSTRIAL IOT & ENERGY MONITORING SYSTEM

*Implementation Lead & Hardware Designer*
2025 – Present

*An end-to-end IIoT solution for real-time monitoring of industrial generators and Rural Electrification Board (REB) power lines. Bridges legacy industrial machinery with cloud-based analytics using LPWAN and standard industrial protocols.*

- Industrial IoT & Energy Monitoring System
- Implementation Lead & Hardware Designer
- An end-to-end IIoT solution for real-time monitoring of industrial generators and Rural Electrification Board (REB) power lines. Bridges legacy industrial machinery with cloud-based analytics using LPWAN and standard industrial protocols.
- Engineered firmware for RAK3173 & ESP32/ESP8266 to fetch telemetry via RS485/RS232 (Modbus RTU/ASCII).
- Led site inspections and machine compatibility testing as primary technical lead.
- Standardized field deployment process; designed rugged industrial enclosures for harsh environments.
- Managed edge-to-server connectivity ensuring high uptime for critical energy data transmission.

### HOSPITAL MANAGEMENT & WORKFLOW SYSTEM

Project Manager · Southern IoT

A comprehensive healthcare digitization platform optimizing clinical workflows and patient data management, scaled from a single-department pilot to a hospital-wide implementation.

- Automated patient admission, discharge, and inter-department transfers with consistent data integrity.
- Built centralized EHR database architecture with HIPAA-aligned security protocols.
- Developed APIs connecting Pharmacy, Pathology, and Radiology departments into a single source of truth.
- Designed a high-density clinical dashboard to reduce cognitive load for medical practitioners.

### CNC ROUTER & LASER CNC MACHINE

Designer & Builder · Southern IoT

Designed and built two CNC machines from scratch — a CNC Router for precise workpiece dimensioning and a Laser CNC for engraving and cutting operations.

### AUTOMATIC WIRE BENDING MACHINE

Designer & Developer · Southern IoT

Developed an automated wire bending machine capable of forming wires into predefined shapes with high repeatability, minimizing manual intervention in production.

### KNITTING FLOOR MONITOR — LIVE MACHINE DATA OVER ETHERNET

Developer · Southern IoT

Decoded the GD NetLink (UDP) status reports of JINHAO flat knitting machines into a listen-only Python monitor with a live web dashboard, tracking run state, speed, cycle time, stops and alarms.

## UNDERGRAD PROJECTS

### EV MOTOR FAULT DETECTION & HEALTH MONITORING SYSTEM

Lead Researcher · Undergraduate Thesis, CUET

Portable multi-sensor system using Raspberry Pi 4 & ESP32 for real-time EV motor monitoring with cloud analytics pipeline and custom PCB design.

**Autonomous Mobile Robot — 3D SLAM & Navigation**
Pre-Defence Project · CUET

ROS 2-based Simulation for autonomous robot for 3D SLAM and navigation, designed as a functional office desk delivery robot.

**INDUSTRIAL OBJECT PICK & PLACE ROBOT**
4-DOF robotic arm with Pi Camera for object detection and automated placement.

**IOT-BASED IPS MONITORING SYSTEM**
Real-time power monitoring and control of IPS units with IoT relay management.

## TRAINING & INDUSTRIAL EXPOSURE

**INDUSTRIAL AUTOMATION COURSE**
IICT, CUET & BCC, ICT Division — 60 Hours

- FluidSIM, TIA Portal, PLC Ladder Logic Programming, HMI Design, Pneumatic & Electro-Pneumatic Systems.

**INDUSTRIAL ATTACHMENT — WALTON HI-TECH INDUSTRIES PLC** — **JAN 2025 · 14 DAYS**
GAZIPUR

- Injection/extrusion molding, PU foaming, thermoforming, PCB & cable production, AC/refrigerator assembly & QC testing.

**BASIC SAFETY TRAINING**
SAMSUNG C&T CORPORATION

- First Aid, Fire Safety, PST, PSSR, Emergency Procedures.

## SOLIDWORKS CERTIFICATION

**SOLIDWORKS 3D CREATOR ASSOCIATE**

**SOLIDWORKS CAD Design Associate**

## REFERENCES

**Sanjeeb Roy**
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**Dr. Homayun Kabir**
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